

The Perturbation Operator

Softmax, the Sigmoid of Fig. 1f, and the Null Sink at One Deletion

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1 Three operators that differ in one step ✗

1.1 Symbols ✗

Table 1 defines every symbol used in this document. Notation follows the manuscript’s Methods.

All three operators share the score, the value, and the residual block that follows,

$$\Delta h_i = \sum_{t=1}^M \beta_{i,t} v_t, \quad \hat{h}_i = \text{LN}(h_i + \Delta h_i), \quad h_i^{\text{pert}} = \text{LN}(\hat{h}_i + \text{FFN}(\hat{h}_i)), \quad (1)$$

where LN is layer normalization and FFN a position-wise feed-forward network. They differ only in how the score becomes the weight $\beta_{i,t}$.

1.2 Operator A: softmax over perturbations ✗

The Methods and the code’s default.

$$\beta_{i,t}^A = \frac{e^{s_{i,t}}}{\sum_{u=1}^M e^{s_{i,u}}}, \quad \sum_{t=1}^M \beta_{i,t}^A = 1. \quad (2)$$

At $M = 1$ the single weight is $e^{s_{i,1}}/e^{s_{i,1}} = 1$ for every gene, whatever the score, so $\Delta h_i = v_1$: one vector shared by all 6,607 genes. The score cancels, and the 16,200 weights of W_Q and W_K receive zero gradient on single-deletion data. Measured in evaluation mode, the largest difference between any gene’s Δh_i and gene 0’s is 0.0. SRC: notes-tex/figure-3-gate/review/2026-10-04/05_architecture_and_operator.md

Every expression, proteome and morphology run used operator A except the arms of section 3. The gene-interaction models of experiments 025 and 030 use it too, at $M = 2$ or 3, where the softmax is not degenerate.

Table 1: Symbols, in alphabetical order. Widths are those of the 1,084,619-parameter expression model (`conf/cgt_expr_012.yaml`); the 6.68 million parameter model of rounds v13 to v20 has the same operator at the same width. SRC: `paper/nature-biotech/sections/methods.tex`; `torchcell/models/equivariant_cell_graph_transformer.py`

symbol	meaning
$a \in [h]$	index of an attention head; the operator has $h = 9$
b_K, b_V	bias vectors of the key and value projections
$b_\emptyset$	the null sink’s one learned scalar (<code>null_bias</code>), shared by all heads
$\beta_{i,t}$	weight gene i puts on perturbation t , one scalar per head
d	width of a gene state, 90
$d_k = d/h$	width of one head, 10
$\delta_{i,t}$	deleted-gene flag: 1 if gene i is the gene that perturbation t deletes, else 0
Δh_i	vector the operator adds to gene i ’s state
$g(t)$	index of the gene that perturbation t deletes
$h_i \in \mathbb{R}^d$	wild-type state of gene i , output of the encoder F_θ ; the same for every strain
h_i^{pert}	state of gene i in the perturbed strain, output of the operator
$k_t = \mathbf{p}_t W_K$	key of perturbation t
M	number of perturbations in a strain; 1 for every expression, proteome and morphology strain used here
N	number of genes, 6,607
$\mathbf{p}_t \in \mathbb{R}^d$	token of perturbation t . Methods: $\mathbf{p}_t = r(e_t) + \mathbf{t}_{\tau_t} + m_t \mathbf{w}$. Code: $\mathbf{p}_t = h_{g(t)}$; the type embedding $\mathbf{t}_\tau$ and the magnitude term $m_t \mathbf{w}$ are not implemented
$q_i = h_i W_Q$	query of gene i
$s_{i,t} = q_i \cdot k_t / \sqrt{d_k}$	score: relevance of perturbation t to gene i
$\sigma(x) = 1/(1 + e^{-x})$	logistic sigmoid
$v_t = \mathbf{p}_t W_V$	value of perturbation t , the content that can be delivered
W_O	output matrix mixing the concatenated heads
W_Q, W_K, W_V	query, key and value projection matrices

1.3 Operator B: the per-pair sigmoid of Fig. 1f ✗

$$\beta_{i,t}^B = \sigma(s_{i,t}) = \sigma\left(\frac{(h_i W_Q) \cdot (\mathbf{p}_t W_K)}{\sqrt{d_k}}\right). \quad (3)$$

No sum over t appears. Each weight is set by its own score and lies in $(0, 1)$. At $M = 1$ the weight differs from gene to gene and W_Q, W_K receive gradient. At $M = 3$ the three weights can all be near 1, so effects add, or all near 0, so the gene ignores all three. Operator B exactly as drawn has not been implemented; the null sink is its closest implemented relative.

1.4 Operator C: the null sink ✗

The null sink (`null_sink=true`) prepends one extra key and value to the perturbation set: an all-zero token, index $\emptyset$, with a learned scalar $b_\emptyset$ added to its logit. The softmax then runs over $M + 1$ entries. The zero token passes through projections that carry biases, so its key is b_K and its value is b_V ; the deleted gene’s key is $\mathbf{p}_t W_K + b_K$, and b_K cancels between the two logits. At $M = 1$,

$$\beta_{i,1}^C = \frac{e^{s_{i,1}}}{e^{s_{i,1}} + e^{b_\emptyset}} = \sigma(s_{i,1} - b_\emptyset), \quad \Delta h_i = \beta_{i,1}^C v_1 + (1 - \beta_{i,1}^C) b_V. \quad (4)$$

A two-way softmax is the sigmoid of the difference of its two logits, which is why the null sink has the form of eq. (3) at one deletion. Table 2 lists where the two part.

Table 2: The null sink equals the operator of Fig. 1f only at one deletion. For multi-gene strains and for natural isolates it keeps the softmax’s shared budget among the real perturbations.

SRC: `torchcell/models/equivariant_cell_graph_transformer.py, null-sinkdesigncomment`

	B (Fig. 1f)	C (null sink)
threshold	none; $\beta = 0.5$ at score 0	$b_\emptyset$, learned, one scalar for all nine heads
what the unused share delivers	nothing	b_V , a learned constant vector
at $M \geq 2$	weights independent; their sum lies between 0 and M	a softmax over $M + 1$ entries; the weights on real perturbations sum to at most 1, so effects are averaged and never added
magnitude	as is	optional rescale by $1/\sigma(-b_\emptyset^{\text{init}})$ (<code>null_sink_magnitude_match</code>), which restores the reference’s norm at initialization

Initialization decides whether the sink takes part. With $b_\emptyset = -4$ the gate is $\sigma(s + 4) \approx 0.98$: the sink holds about 2 percent of the mass and the operator is nearly operator A. In the first attempt the bias moved from -4.0 to between

−3.92 and −3.95 by epoch 187. With $b_{\mathcal{O}} = 0$ the gate starts at 0.5, and the model has to close the sink actively for it to be useless. SRC: [notes/experiments.019-simb-multimodal.scripts.gh_expr_008_arm.md](#)

1.5 Which normalization makes the weights uniform ✗

Two normalizations are in play and they behave differently.

Normalizing the weights over perturbations restores the degeneracy. If the sigmoid weights are divided by their sum over t , $\tilde{\beta}_{i,t} = \beta_{i,t}^B / \sum_u \beta_{i,u}^B$, then at $M = 1$ $\tilde{\beta}_{i,1} = 1$ again. Any normalization across the perturbation axis reproduces operator A at one deletion. Fig. 1f as drawn has none.

The layer normalization that follows does not. LN acts on each gene’s vector separately, across its d coordinates, and does not compare genes. Two genes with $\beta_i = 0.1$ and $\beta_j = 0.9$ receive different mixtures of their own state and v , with different directions, and LN preserves direction. It removes overall magnitude: if $\beta_i v$ is far larger than h_i for every gene, every $\hat{h}_i$ approaches the direction of v and the genes become alike. That is a matter of scale, $\|v\|$ against $\|h_i\|$, and not a degeneracy.

2 What a gene can choose ✗

2.1 How much against what ✗

With h heads, the vector delivered at $M = 1$ under operator B or C is

$$\Delta h_i = \left[\beta_i^{(1)} v^{(1)} \parallel \dots \parallel \beta_i^{(h)} v^{(h)} \right] W_O = \sum_{a=1}^h \beta_i^{(a)} u^{(a)}, \quad u^{(a)} = [0 \parallel \dots \parallel v^{(a)} \parallel \dots \parallel 0] W_O, \quad (5)$$

where $v^{(a)} \in \mathbb{R}^{d_k}$ is head a ’s slice of the value and $u^{(a)} \in \mathbb{R}^d$ is that slice after output mixing. The h vectors $u^{(a)}$ depend on the deleted gene alone. Gene i contributes the h scalars $\beta_i^{(a)} \in (0, 1)$ and nothing else.

How much. Gene i sets nine dials, each between 0 and 1, each on a fixed vector.

What. The content of each $u^{(a)}$, which gene i cannot change.

Over all genes Δh_i therefore lies in the span of nine vectors with bounded non-negative coefficients, and the gene-dependent part of the delivered signal has rank at most $h = 9$ of $d = 90$. Measured at initialization with the sink open, in six draws: rank exactly 9, carrying 4.75 to 5.90 percent of the context energy; with the sink off, rank 0 to floating-point precision. SRC: [notes/experiments.019-simb-multimodal.multiplicative-perturbation-conditioning.md, section1.3](#)

For a gene to choose what it receives, the delivered vector has to be a function of both gene and perturbation with rank up to d . Three forms, by increasing capacity:

$$\text{diagonal multiplicative: } \Delta h_i = h_i \odot \gamma(\mathbf{p}_t) \quad \text{rank } d, \quad (6)$$

$$\text{low-rank bilinear: } \Delta h_i = \sum_{r=1}^R (a_r^\top h_i) (c_r^\top \mathbf{p}_t) o_r \quad \text{rank } R, \quad (7)$$

$$\text{gate plus product: } \Delta h_i = \sigma(s_{i,t}) v_t + h_i \odot \gamma(\mathbf{p}_t). \quad (8)$$

Here $\odot$ is the elementwise product, $\gamma : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a small learned map from the perturbation token to one multiplier per channel, and $a_r, c_r, o_r \in \mathbb{R}^d$ are the learned vectors of the r -th bilinear component. In eq. (6) each of gene i ’s 90 channels is scaled by a deletion-specific factor, so two genes with different states receive different vectors, not different amounts of one vector.

Operator A is not without gene-specific response. The layers after it apply a nonlinear function to the sum $h_i + v$, and the difference between two strains’ predicted profiles has standard deviation 0.061 across reporters against 0.163 for a profile itself. The interaction exists as the curvature of one shared function of $h_i + v$. SRC: [notes-tex/figure-3-gate/review/2026-10-04/05_architecture_and_operator.md](#)

2.2 The deleted-gene flag ✗

The flag $\delta_{i,t}$ enters as one more input to the readout, or as a learned vector $\mathbf{e}_{\text{self}}$ added to that one token, $\hat{h}_i \leftarrow \hat{h}_i + \delta_{i,t} \mathbf{e}_{\text{self}}$.

Why it is needed. Under operator A every gene receives the same v_t , the deleted gene included. The model can recognize the deleted gene only through the coincidence that its own state equals the source of the key. The model predicts +0.036 for the deleted gene’s own transcript against a measured mean of -2.42 . SRC: [notes-tex/figure-3-gate/review/2026-10-04/07_data_splits_metrics.md](#)

What it buys. One entry per strain, the largest and most predictable. A predictor that writes only that entry scores 0.0175 per-feature Pearson, which bounds the flag’s value on its own. It carries no information about the other 6,606 genes. Hypothesis (untested): under a sigmoid gate the model can learn a large self-score $s_{g(t),t}$ without the flag, since query and key then come from the same gene. When models are compared the deleted gene’s own entry should be blanked in scoring whether or not a flag is used.

3 What has been measured ✗

The null sink has four attempts on file (table 3). The audit of 2026-10-04 reported one, stopped before epoch 190; that was incomplete.

Table 3: The sigmoid gate has one run at a resolving budget, on one seed. All four attempts use the 1,084,619-parameter expression model on split seed 0. Scores are validation per-feature Pearson by rolling maximum, an upward-biased order statistic. Across-initialization standard deviation on a fixed split is about 0.006, so the last row’s difference of -0.010 on one seed is not resolved. SRC: [experiments/019-simb-multimodal/results/round_leaderboards.csv](#); [results/decoder_arms_torchcell_019_expr_v8.csv](#); [results/wave_convergence_stage-wave4b.csv](#)

arm	$b_\emptyset$ at init	matched	epochs	seeds	W&B run (seed 0)	result
A4_nullsink	-4	no	186 to 189, crashed	4	vrhntd2e	no co-launched reference
N_sink0	-4	no	78 to 80, crashed	4	3eugl3dq	none; the bias barely moved
N_sink_mm	0	yes	300	4 pairs	v1grwe46	+0.0024, sd 0.0090, against R_ref
V_sink	0	yes	4,184 (48 h)	1 pair	fqk5mbr3	0.200 against 0.210 for V_ref (sm6efleg)

The same round ran the higher-rank forms once each at the same budget (table 4).

Table 4: One run of each pair mechanism at about 4,100 epochs. Pair rank is the rank of the gene-dependent signal the mechanism can deliver at one deletion. One seed each on split seed 0, so no row is a replicated result and the ordering is not established. Scores as in table 3. SRC: [experiments/019-simb-multimodal/results/round_leaderboards.csv](#), [rowstagedstage-wave6](#)

arm	mechanism	pair rank	epochs	score	W&B run
V_ref	operator A	0	4,184	0.210	sm6efleg
V_sink	null sink, open, magnitude matched	9	4,184	0.200	fqk5mbr3
V_basis16	rank-16 response basis at the readout	16	4,184	0.204	hoheq7rz
V_basis32	rank-32 response basis	32	4,109	0.193	gykwmhro
V_basis64	rank-64 response basis	64	4,105	0.228	b50f93ju
V_film	elementwise product at the readout	90	4,105	0.205	1j0tn0p7
V_hadamard	elementwise product at the operator, eq. (6)	90	4,091	0.214	y69alttye
V_hadamard_add	product added to the additive context, as eq. (8) with operator A’s first term	90	4,091	0.215	9ahs039r

At a real budget the sigmoid gate alone showed no gain in its one run. That is weak evidence, and it agrees with section 2: nine dials are a small change. Not yet run: the gate together with a rank- d product (eq. (8) in full), either with the flag, and any of these on more than one seed or split.

Run pages, https://wandb.ai/zhao-group/torchcell_019_expr_v8/runs/fqk5mbr3 for the long null sink run and https://wandb.ai/zhao-group/torchcell_019_expr_v8/runs/sm6efleg for its reference.

4 Other models and other regimes ✗

4.1 Published perturbation models ✗

Table 5 places the three operators beside published designs. The published rows repeat the reading recorded in [notes-tex/019-simb-multimodal](#), section “What the perturbation-response literature says”; they are descriptions of other groups’ models, and none was measured here. In that table h_u is the other model’s gene state and $h^{\mathcal{P}}$ its perturbation embedding. The published models are GEARS  [external: Roohani 2023, doi:10.1038/s41587-023-01905-6],

Table 5: Published models are additive where the perturbation enters and gene-specific at the readout. The State row has no entry for combining several perturbations because the earlier reading did not record it and the paper could not be re-read on 2026-10-05. SRC: notes-tex/019-simb-multimodal/sections/10-launch.tex, tab:related-operators

model	tokens	where the perturbation enters	several tions	perturba-	source of gene-specific response
CGT, operator A	genes	after the encoder, $\text{LN}(h_i + \sum_t \beta_{i,t} v_t)$, softmax	weighted average		shared FFN and one shared readout MLP
CGT, Fig. 1f	genes	after the encoder, sigmoid gate	gated sum		nine dials per gene, then the same shared layers
CGT, 025 and 030	genes	operator A, with the CLS token as one more query	weighted average		scalar readout from the perturbed CLS; $M \geq 2$, so the softmax selects
GEARS	genes	$\text{MLP}(h_u + h^{\mathcal{P}})$, with $h^{\mathcal{P}}$ propagated over a gene graph	sum of embeddings		per-gene output layer $w_u \in \mathbb{R}^d$, b_u
State	cells	$\mathbf{H}_{\text{cell}} + \mathbf{H}_{\text{pert}} + \mathbf{H}_{\text{batch}}$ into a transformer over cells	not recorded		readout $\mathbf{W}_{\text{recon}} \in \mathbb{R}^{d_h \times G}$, one column per gene
CPA	cells	additive in latent space	sum		decoder
token edit	genes	before the encoder: the gene’s token is removed or flagged	several edited	tokens	the encoder’s self-attention carries the edit to every gene

State▲^[external: Adduri 2025, doi:10.1101/2025.06.26.661135], CPA▲^[external: Lotfollahi 2023], and the token-edit family, Geneformer and scGPT▲^[external: Theodoris 2023; Cui 2024].

The difference from GEARS and State is the readout. Both are additive at the point of entry, as operator A is. Each gives every gene its own output parameters. CGT reads every gene through one shared MLP, so gene identity enters only through h_i . A per-gene output weight in the style of GEARS costs $6,127 \times 91 = 557,557$ parameters on the expression model.

Token editing avoids the one-key problem and forfeits the single encode. Each gene attends to the edited gene among 6,607 tokens, so no key set of size one arises. The cost is one encoder pass per strain.

The trigenic result and the single-deletion result are different regimes. Hypothesis (untested): operator A serves experiments 025 and 030 because a trigenic interaction is one scalar per strain at $M = 3$, where the softmax chooses among keys and the CLS summarizes; expression asks for about 6,000 gene-specific values at $M = 1$, where the same operator delivers one vector to all genes. The trigenic state of the art supports the operator in its regime and is silent on the single-deletion, per-gene regime.

4.2 Epistasis forms at two or more perturbations ✕

Table 6 gives the null models used for the free fatty acid study, with f a phenotype normalized to the reference strain, f_i a single mutant and f_{ij} a double.

Table 6: Each null model defines interaction as the departure from a different expectation. $x_i \in \{0, 1\}$ marks the presence of mutation i , and $\beta_0, \beta_i, \beta_{ij}$ in the last row are regression coefficients, unrelated to the attention weights $\beta_{i,t}$.

SRC: notes/experiments.008-xue-ffa.results.002-interaction-models-report.md

null model	expectation without interaction	interaction term
additive	$f_{ij} = f_i + f_j - 1$	$\varepsilon_{ij}^{\text{add}} = f_{ij} - f_i - f_j + 1$
multiplicative	$f_{ij} = f_i f_j$	$\varepsilon_{ij}^{\text{mult}} = f_{ij} - f_i f_j$
log-additive	$\log f_{ij} = \log f_i + \log f_j$	$\varepsilon_{ij}^{\text{log}} = \log f_{ij} - \log f_i - \log f_j$
GLM, log link	$\log \mathbb{E}[f] = \beta_0 + \sum_i \beta_i x_i$	coefficients β_{ij} of the products $x_i x_j$

The following is reasoning from the equations; none of it is measured.

Operator A averages. $\Delta h_i = \sum_t \beta_{i,t} v_t$ with weights summing to 1, so a double mutant’s shift is a convex combination of the two single shifts and never their sum. Neither the additive nor the multiplicative null is the operator’s default, and the layers after it have to learn even the case of no interaction.

Operator B sums. $\Delta h_i = \sum_t \sigma(s_{i,t}) v_t$, so the latent shift of a double is the sum of the single shifts. With a linear readout that is the additive null; with a readout that predicts $\log f$ it is the log-additive null. Interaction is then what the model adds beyond the sum.

An explicit second-order term names the interaction. With $t < u$ ranging over pairs of perturbations in the

strain and ϕ a small learned map,

$$\Delta h_i = \underbrace{\sum_t \beta_{i,t} v_t}_{\text{single effects}} + \underbrace{\sum_{t < u} \beta_{i,t} \beta_{i,u} \phi(v_t \odot v_u)}_{\text{pairwise interaction as seen by gene } i} \quad . \quad (9)$$

The second sum is empty at $M = 1$, so it costs nothing on single-deletion data and cannot be learned from it.

Single deletions can teach the first-order map: how deleting gene $g(t)$ moves every other gene i , the array of $\beta_{i,t} v_t$ over all pairs (i, t) . Sections 1 and 2 concern that object. The pairwise term needs double and triple mutants: the Sameith doubles for expression, and the stores of experiments 025 and 030 for fitness.